

팍리스 실무형 일경험 프로그램

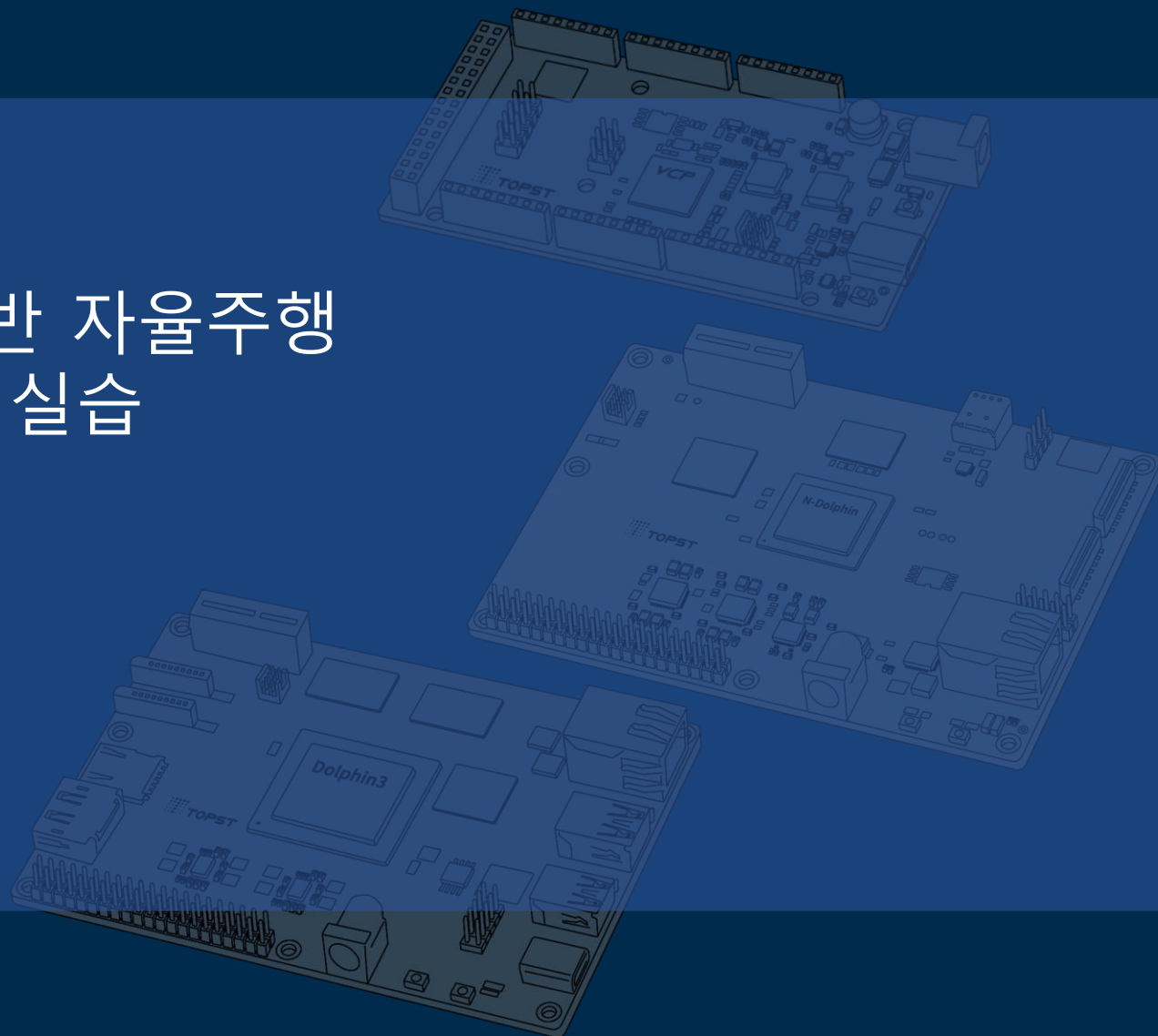
SDV 아키텍처 기반 자율주행 통합 시스템 구현 실습



6일차

01 AI-G 환경 구성

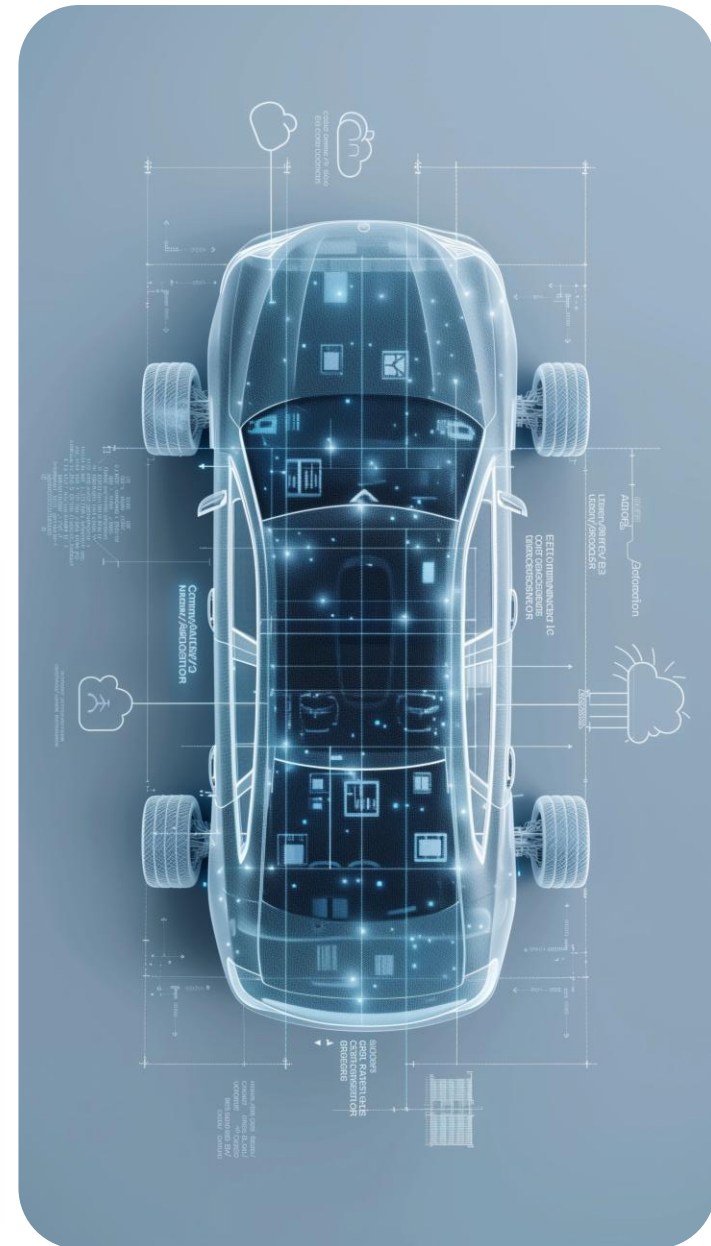
Telechips TOPST



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AI-G Specification

Part Name	AI-G
CPU	Arm® Cortex-A53 1.45 GHz, Quad
NPU	1.0 Ghz, 8 TOPS
Memory	2 GB 32-bit LPDDR4X
Storage	8 GB eMMC 5.1 Flash
Network	Gigabit Ethernet
Camera	MIPI CSI-2 2-lane, MIPI CSI-2 4-lane
Display	MIPI DSI-2 2-lane
Button	System reset button, Boot mode button
Others	PCIe Gen3, FWD debugger port, I2S, I2C, SPI, CAN, JTAG
Power	5V/3A
OS	Linux
Mechanical	120 mm x 90 mm

- AI-G 는 1개의 CPU와 1개의 NPU 존재
 - Cortex-A53
 - NPU : 8TOPS (4TOPS Dual)
 - 2GB RAM
 - 8GB eMMC
 - Ethernet, DSI, CSI, PCIe,

AI-G Yocto

Build Source 가져오기 (WSL Ubuntu)

```
$ cd ~/Work  
$ mkdir ai-g-yocto  
$ cd ai-g-yocto  
$ repo init -u https://github.com/topst-development/manifests.git -b develop -m linux_yp4.0_topst.xml
```

```
ead@TC-D-20-0546:~/topst-sdk$ repo init -u https://github.com/topst-development/manifests.git -b develop -m  
linux_yp4.0_topst.xml  
Downloading Repo source from https://gerrit.googlesource.com/git-repo  
  
... A new version of repo (2.54) is available.  
... New version is available at: /home/ead/topst-sdk/.repo/repo/repo  
... The launcher is run from: /usr/bin/repo  
!!! The launcher is not writable. Please talk to your sysadmin or distro  
!!! to get an update installed.  
  
Your identity is: Topst Developer <topstdeveloper@gmail.com>  
If you want to change this, please re-run 'repo init' with --config-name  
  
repo has been initialized in /home/ead/topst-sdk
```

AI-G Yocto

```
$ repo sync
```

```
ead@TC-D-20-0546:~/topst-sdk$ repo sync
```

```
... A new version of repo (2.54) is available.  
... New version is available at: /home/ead/topst-sdk/.repo/repo/repo  
... The launcher is run from: /usr/bin/repo  
!!! The launcher is not writable. Please talk to your sysadmin or distro  
!!! to get an update installed.
```

```
Syncing: 25% (3/12), done in 35.713s
```

```
Syncing: 100% (12/12) 0:52 | 1 job | 0:14 meta-openembedded @ poky/meta-openembeddedrepo sync has finished  
successfully.
```

```
$ ls
```

```
ead@TC-D-20-0546:~/topst-sdk$ ls
```

```
easy-setup.sh  mktcing  poky  stitch-fai-ai.sh  stitch-fai-d3.sh  tools
```

AI-G Yocto

topst-build.sh 실행

```
$ ./easy-setup.sh
```

스크롤 내리기 -> 방향키(->) 클릭 -> 엔터 -> accept 엔터

```
End User License Agreement

# Open source License Agreement
Welcome to TOPST.AI

We are pleased to provide you access to various open source software ("Software") on our website. Before you proceed to
use any of the Software, we kindly request that you carefully read and understand the following declaration.

## Open source License Agreement
All Software provided on our website is subject to the terms and conditions of the respective open source licenses
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## Permitted Use
You are granted a non-exclusive, worldwide, royalty-free license to use, modify, and distribute the Software in
accordance with the terms of the applicable open source licenses. Please be aware that each open source license may
have specific requirements, such as attribution, sharing derivative works under the same license, and providing the
source code when distributing modified versions. Make sure to review and adhere to the requirements of each license.

## No Warranty
The Software is provided on an "AS IS" basis, without warranties or conditions of any kind, either expressed or

<Proceed to confirm>
```



AI-G Yocto

topst-build.sh 실행

\$ source poky/meta-topst/topst-build.sh -> 1번 머신 선택 (1 입력)

```
ead@TC-D-20-0546:~/topst-sdk$ source poky/meta-topst/topst-build.sh
Choose MACHINE
  1. ai-g-topst
  2. d3-g-topst-main
  3. d3-g-topst-sub
  4. d5-g-topst-main
  5. d5-g-topst-sub
select number(1-5) => 1
machine(ai-g-topst) selected.
You had no conf/local.conf file. This configuration file has therefore been
created for you from /home/ead/topst-sdk/poky/meta-topst/template/ai-g-topst/local.conf.sample
You may wish to edit it to, for example, select a different MACHINE (target
hardware). See conf/local.conf for more information as common configuration
options are commented.
```

```
ead@TC-D-20-0546:~/topst-sdk/build/ai-g-topst$ ls
bitbake-cookerdaemon.log  buildhistory  cache  conf  downloads  sstate-cache  tmp
```

AI-G Yocto

쓰레드 개수 조정하기

현재는 쓰레드 수가 8로 제한되어 있기 때문에 제공하는 pc의 쓰레드를 최대(12)로 사용하기 위해 수정

```
$ vi conf/local.conf
```

```
# This file is your local configuration file and is where all local user settings
# are placed. The comments in this file give some guide to the options a new user
# to the system might want to change but pretty much any configuration option can
# be set in this file. More adventurous users can look at
# local.conf.sample.extended which contains other examples of configuration which
# can be placed in this file but new users likely won't need any of them
# initially.
#
# Lines starting with the '#' character are commented out and in some cases the
# default values are provided as comments to show people example syntax. Enabling
# the option is a question of removing the # character and making any change to the
# variable as required.

# Parallelism Options
#
# These two options control how much parallelism BitBake should use. The first
# option determines how many tasks bitbake should run in parallel:
#
BB_NUMBER_THREADS ?= "8"
#
# Default to setting automatically based on cpu count
#BB_NUMBER_THREADS ?= "${@bb.utils.cpu_count()}"
#
# The second option controls how many processes make should run in parallel when
# running compile tasks:
#
PARALLEL_MAKE ?= "-j 16"
```


AI-G Yocto

쓰레드 개수 조정

- BB_NUMBER_THREADS ?= "8"을 주석 처리 -> 앞에 # 추가
- a키를 누른 후 수정 가능

```
# This file is your local configuration file and is where all local user settings
# are placed. The comments in this file give some guide to the options a new user
# to the system might want to change but pretty much any configuration option can
# be set in this file. More adventurous users can look at
# local.conf.sample.extended which contains other examples of configuration which
# can be placed in this file but new users likely won't need any of them
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```

AI-G Yocto

캐시 서버 연동

사전에 빌드된 yocto 캐시 서버에서 빌드된 소스들을 가져옴

local.conf 하단에 아래 내용 추가 (외부 IP : 강사 IP / 서비스 포트 : 8080)

INHERIT:remove = "rm_work" 주석해제 -> 빌드 완료 후 작업 파일을 삭제하지 않도록 설정

```
TOPST_FEATURES += "camera"
TOPST_FEATURES += "display"
TOPST_CAM_MODULE += "ov5647"
#TOPST_CAM_MODULE += "imx219"

#DISTRO_FEATURES:append = "pulseaudio"
#DISTRO_FEATURES:remove = "wayland"

#RM_WORK_EXCLUDE += ""
INHERIT:remove = "rm_work"
#OE_TERMINAL = "tmux-new-window"

# Set maximum system partition size. MB
#SYSTEM_PARTITION_SIZE = "5120"

# Enable Network features in U-BOOT
IMAGE_FEATURES += "uboot-net"
```

```
PREMIRRORS:prepend = "\
git://.*.* http://${외부IP}:${서비스포트}/downloads/git2/\n \
ftp://.*.* http://${외부IP}:${서비스포트}/downloads/\n \
http://.*.* http://${외부IP}:${서비스포트}/downloads/\n \
https://.*.* http://${외부IP}:${서비스포트}/downloads/\n \
"

SSTATE_MIRRORS = "\
file://.* http://${외부IP}:${서비스포트}/sstate-cache/PATH \
```

```
PREMIRRORS:prepend = "\
git://.*.* http://${외부IP}:${서비스포트}/downloads/git2/\n \
ftp://.*.* http://${외부IP}:${서비스포트}/downloads/\n \
http://.*.* http://${외부IP}:${서비스포트}/downloads/\n \
https://.*.* http://${외부IP}:${서비스포트}/downloads/\n \
"

SSTATE_MIRRORS = "\
file://.* http://${외부IP}:${서비스포트}/sstate-cache/PATH \
```

AI-G Yocto

저장

- 수정 후 저장하는 법 - esc -> : 입력-> wq 입력 -> 엔터

AI-G tc-nn-app 빌드

tc-nn-app 빌드 수행

- tc-nn-app은 AI-G의 NPU를 사용하기 위한 추론 용 어플리케이션
- 카메라나 이미지를 입력으로 받아 추론 결과를 디스플레이나 이미지 형태로 제공
- 이후 강의에서 tc-nn-app을 통해 차선 인식을 수행할 예정
- 기존의 tc-nn-app에서는 차선 인식 기능이 추가되어 있지 않아 앱 빌드 후 코드 수정 필요

AI-G tc-nn-app 빌드

tc-nn-app 빌드 수행

```
$ bitbake -c compile -f tc-nn-app
```

```

ead@TC-D-20-0546:~/topst-sdk/build/ai-g-topst$ bitbake -c compile -f tc-nn-app
Loading cache: 100% |#####| Time: 0:00:00
Loaded 3867 entries from dependency cache.
Parsing recipes: 100% |#####| Time: 0:00:01
Parsing of 2473 .bb files complete (2460 cached, 13 parsed). 3881 targets, 318 skipped, 0 masked, 0 errors.
WARNING: No recipes in default available for:
/home/ead/topst-sdk/poky/meta-topst/recipes-browser/cog/cog_0.16.bbappend
NOTE: Resolving any missing task queue dependencies

Build Configuration:
BB_VERSION      = "2.0.0"
BUILD_SYS       = "x86_64-linux"
NATIVELSBSTRING = "universal"
TARGET_SYS      = "aarch64-telechips-linux"

Initialising tasks: 100% |#####| Time: 0:00:02
Checking sstate mirror object availability: 100% |#####| Time: 0:00:04
Sstate summary: Wanted 506 Local 164 Mirrors 0 Missed 342 Current 1017 (32% match, 77% complete)
NOTE: Executing Tasks
NOTE: Tasks Summary: Attempted 3899 tasks of which 3226 didn't need to be rerun and all succeeded.
NOTE: Writing buildhistory
NOTE: Writing buildhistory took: 3 seconds

Summary: There was 1 WARNING message.
```

AI-G tc-nn-app 빌드

Yocto 빌드 결과 dir 설명

- Kernel : ai-g-topst-tc-boot-5.10.223-r0.img, Image--5.10.223-r0-ai-g-topst-20251030223228.bin
- Kernel Modules : modules--5.10.223-r0-ai-g-topst-20251030223228.tgz
- Device tree bin : tcc7500-lpd4x321--5.10.223-r0-ai-g-topst-20251030223228.dtb
- Rootfs : telechips-topst-image-ai-g-topst-20251030223228.rootfs.ext4

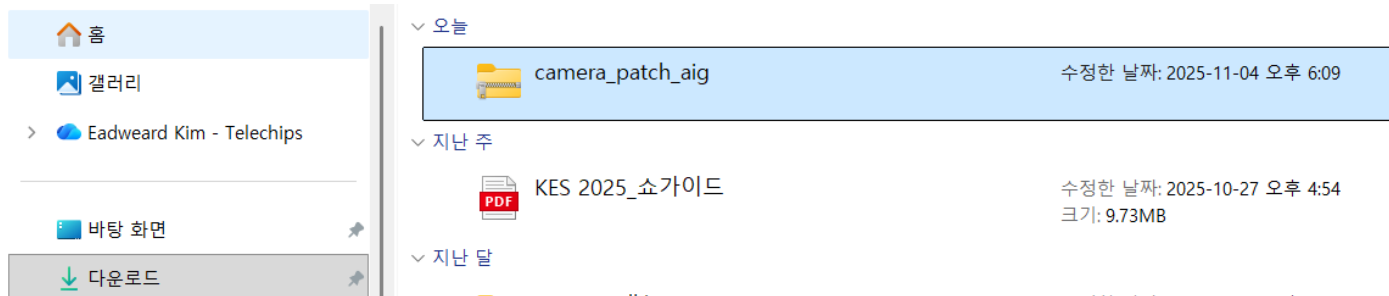
```
ead@TC-D-20-0546:~/topst-sdk/build/ai-g-topst/tmp/deploy/fwdn$ ls
boot-firmware  partition.list  SD_Data.fai  SD_Data.gpt  SD_Data.gpt.back  SD_Data.gpt.prim
```

```
ead@TC-D-20-0546:~/topst-sdk/build/ai-g-topst/tmp/deploy/images/ai-g-topst$ ls
ai-g-topst-tc-boot-5.10.223-r0.img          tc-boot.img
boot-firmware                             tcc7500-lpd4x321--5.10.223-r0-ai-g-topst-20251030223228.dtb
Image                                     tcc7500-lpd4x321-ai-g-topst.dtb
Image--5.10.223-r0-ai-g-topst-20251030223228.bin  tcc7500-lpd4x321.dtb
Image-ai-g-topst.bin                      telechips-topst-ai-image-ai-g-topst-20251030223228.rootfs.ext4
Image-initramfs--5.10.223-r0-ai-g-topst-20251030223228.bin  telechips-topst-ai-image-ai-g-topst-20251030223228.rootfs.manifest
Image-initramfs-ai-g-topst.bin             telechips-topst-ai-image-ai-g-topst-20251030223228.testdata.json
initramfs-topst-image-ai-g-topst-20251030223228.rootfs.cpio  telechips-topst-ai-image-ai-g-topst.ext4
initramfs-topst-image-ai-g-topst-20251030223228.rootfs.manifest  telechips-topst-ai-image-ai-g-topst.manifest
initramfs-topst-image-ai-g-topst-20251030223228.testdata.json  telechips-topst-ai-image-ai-g-topst.testdata.json
initramfs-topst-image-ai-g-topst.cpio      u-boot-ai-g-topst-1.0-r0.rom
initramfs-topst-image-ai-g-topst.manifest  u-boot-ai-g-topst.rom
initramfs-topst-image-ai-g-topst.testdata.json  u-boot.rom
modules--5.10.223-r0-ai-g-topst-20251030223228.tgz  u-boot-topst-initial-env
modules-ai-g-topst.tgz                          u-boot-topst-initial-env-ai-g-topst
tc-boot-ai-g-topst.img                        u-boot-topst-initial-env-ai-g-topst-1.0-r0
```

AI-G 환경 설정

AI-G 이미지 다운 : [AI-G Yocto image](#)

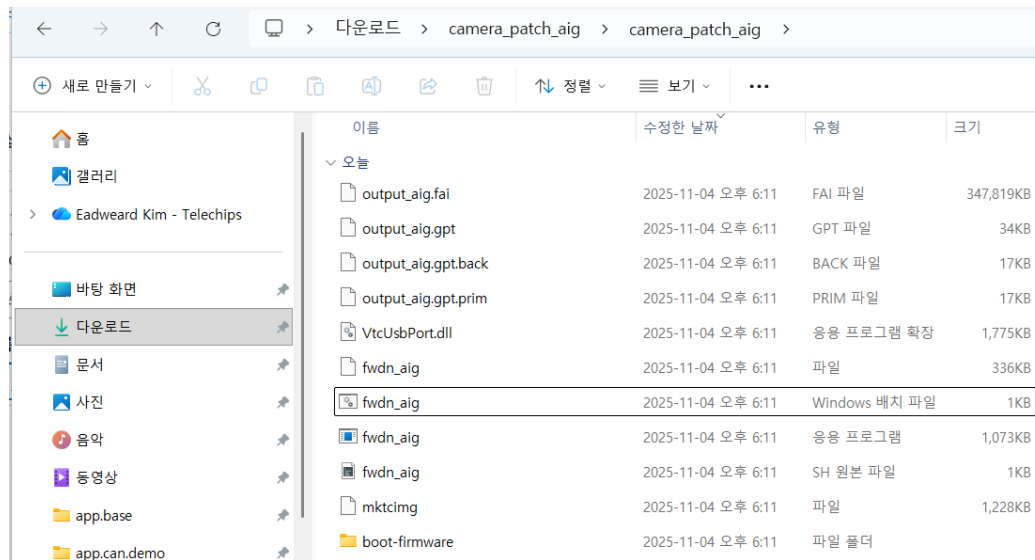
설치 후 다운로드 탭에서 해당 파일 압축 해제



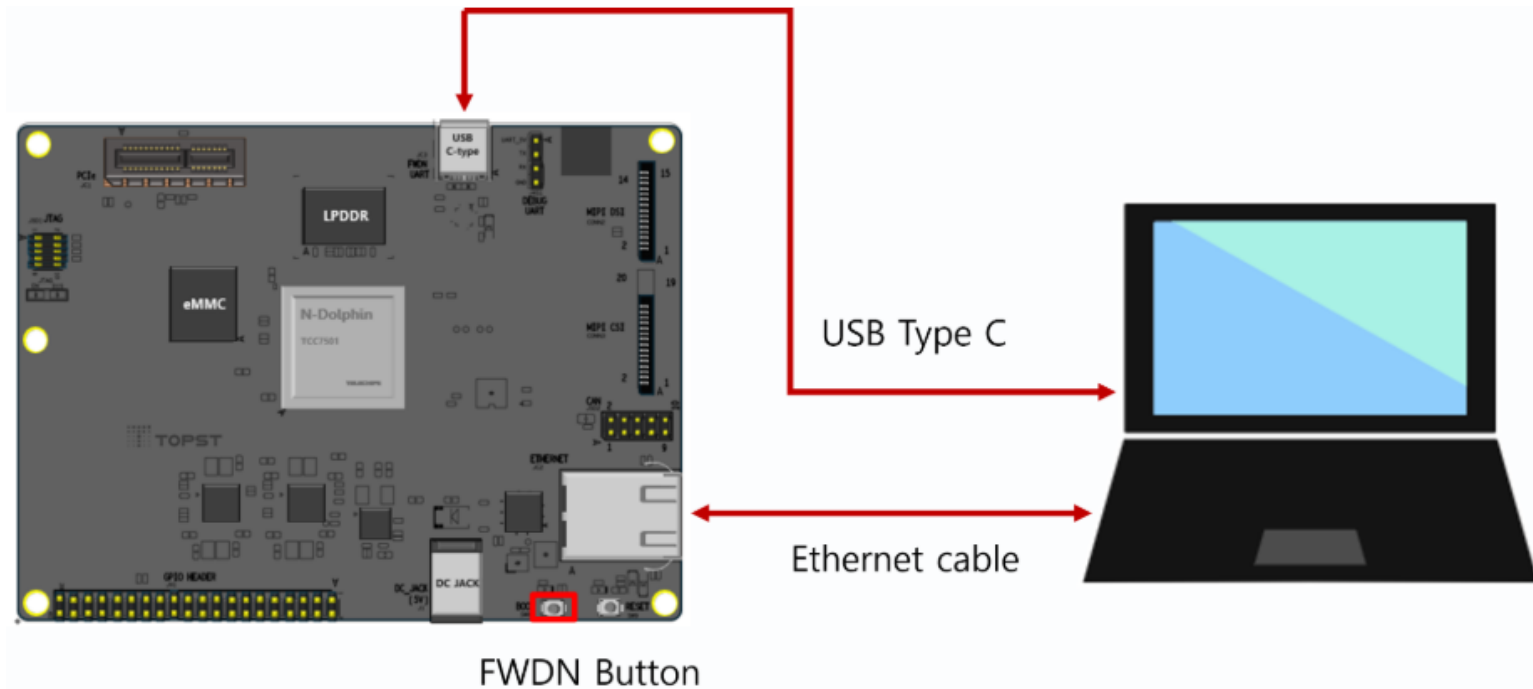
AI-G 환경 설정

AI-G 이미지 다운 : [AI-G Yocto image](#)

fwdn_aig 배치파일 존재하는지 확인



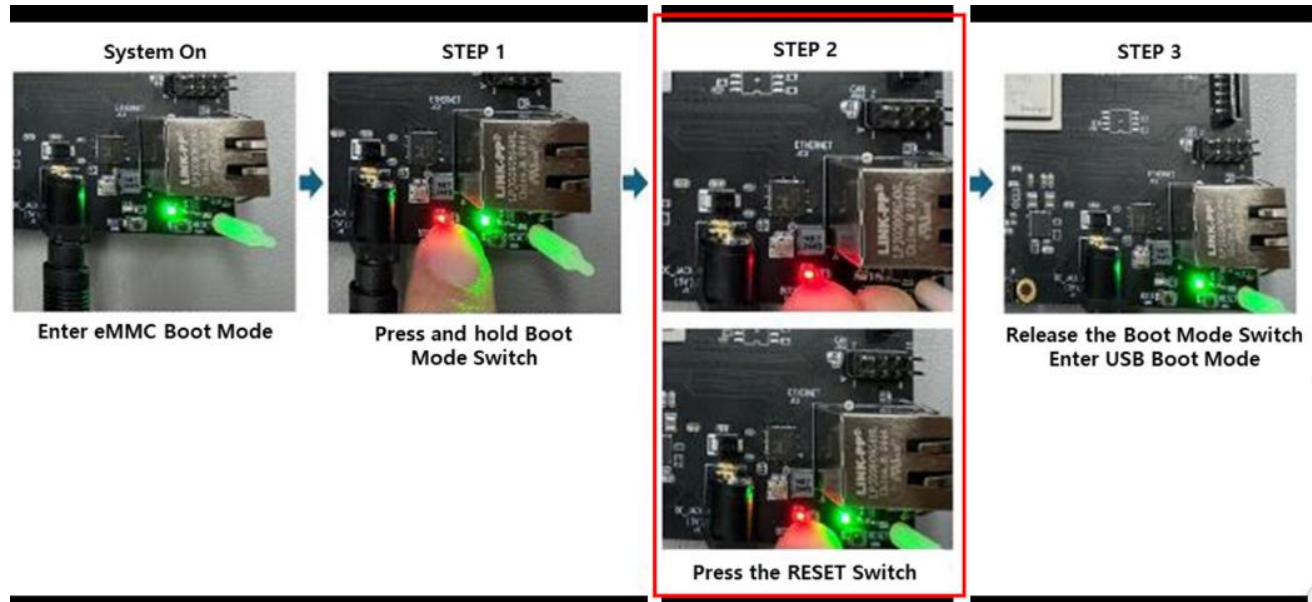
AI-G 환경 설정



- **Connect AI-G Board and Host PC with USB Boot Mode**

- VTC 드라이버가 PC에 정상적으로 설치되어 있는지 확인
- USB Type-C 케이블을 Host PC 및 AI-G의 USB Type-C FWDN 포트에 연결
- 이더넷 케이블을 AI-G 보드의 이더넷 포트와 Host PC에 연결
- FWDN 스위치를 누른 상태에서 전원 케이블을 AI-G 보드에 연결(D3-G보드와 동일)

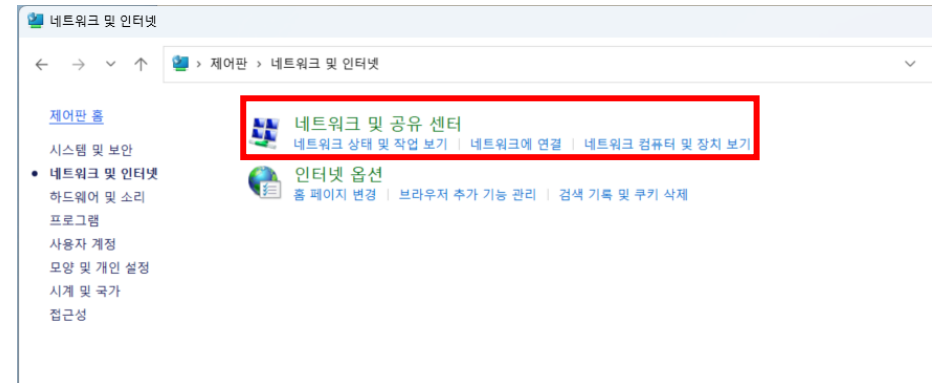
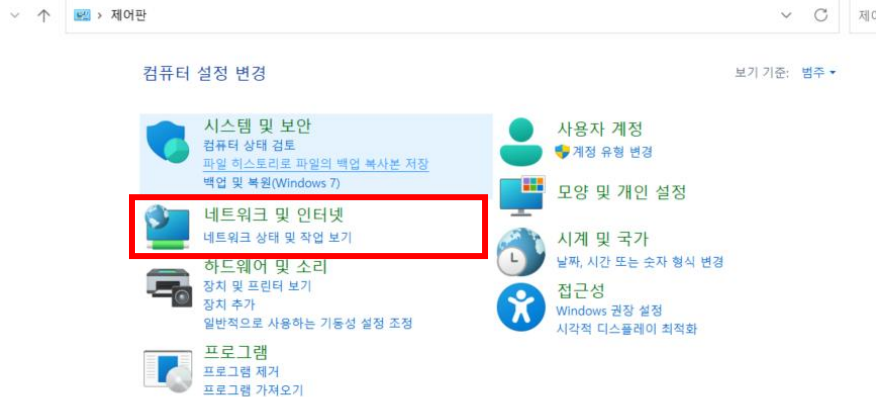
AI-G 환경 설정



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- USB Type-C 케이블을 Host PC 및 AI-G의 USB Type-C FWDN 포트에 연결
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- FWDN 스위치를 누른 상태에서 전원 케이블을 AI-G 보드에 연결(D3-G보드와 동일)

AI-G 환경 설정

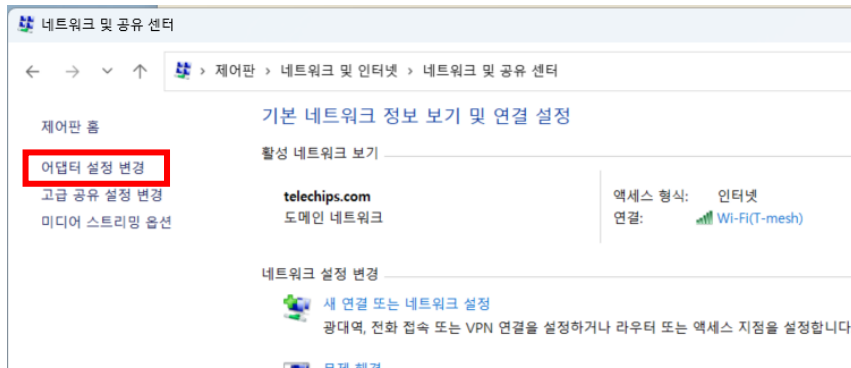


• Ethernet Set Up

▪ Host PC Network Configuration

➤ 제어판 -> 네트워크 및 인터넷 -> 네트워크 및 공유 센터 -> 어댑터 설정 변경

AI-G 환경 설정

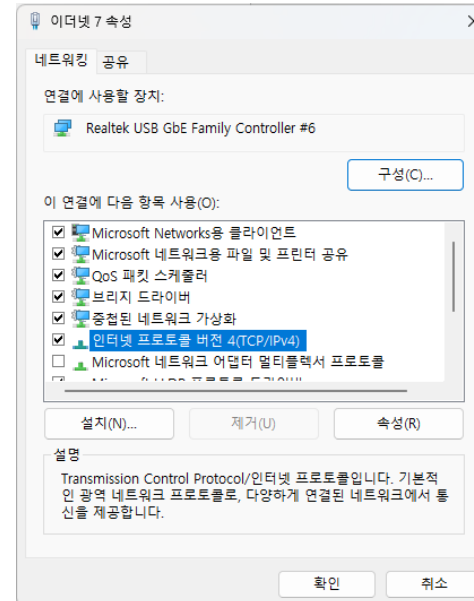


• Ethernet Set Up

▪ Host PC Network Configuration

➤ 제어판 -> 네트워크 및 인터넷 -> 네트워크 및 공유 센터 -> 어댑터 설정 변경

AI-G 환경 설정

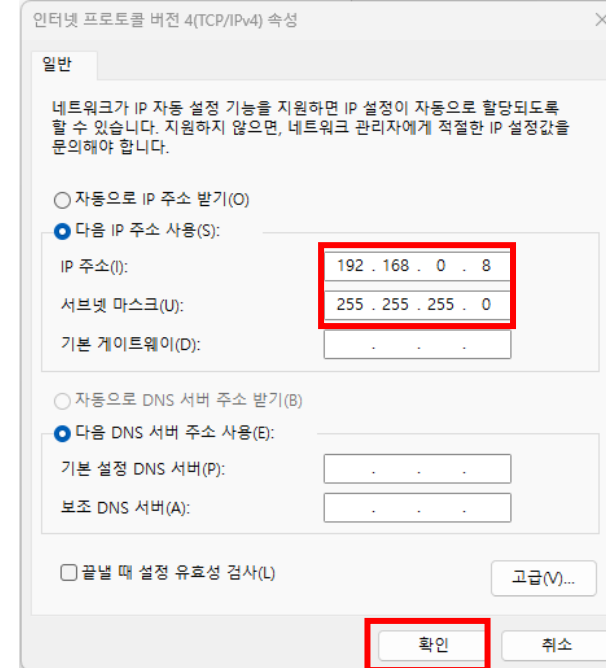
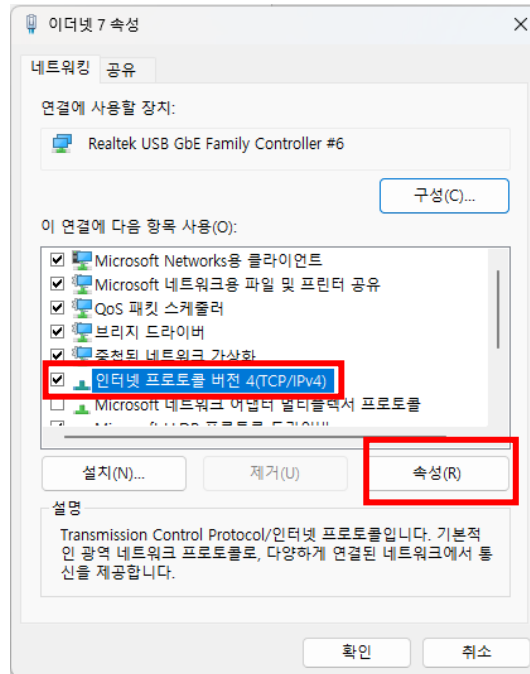


• Ethernet Set Up

▪ Host PC Network Configuration

- 이더넷 선택 -> 속성 -> 인터넷 프로토콜 버전4(TCP/IPv4) -> 속성

AI-G 환경 설정

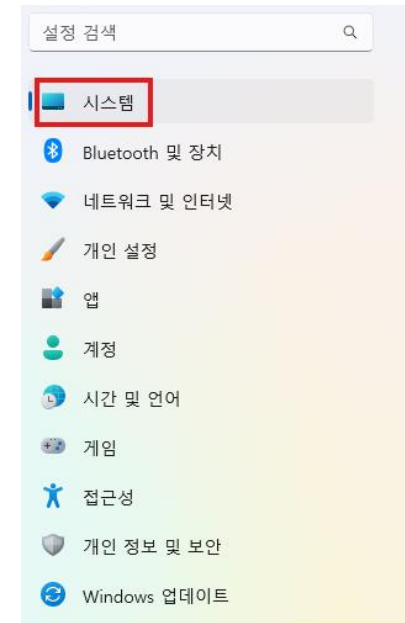


• Ethernet Set Up

▪ Host PC Network Configuration

- 다음 IP 주소 사용 -> IP 주소 : 192.168.0.8 / 서브넷 마스크 : 255.255.255.0 -> 확인

AI-G 환경 설정



- **Add WMIC (설치되어 있지 않은 경우 만)**

- FWDN 전 AI-G 보드의 FWDN 포트에 연결된 포트를 알기 위해 WMIC 설치 필요
 - 시작 메뉴에서 설정 열기
 - 시스템 메뉴로 이동

AI-G 환경 설정

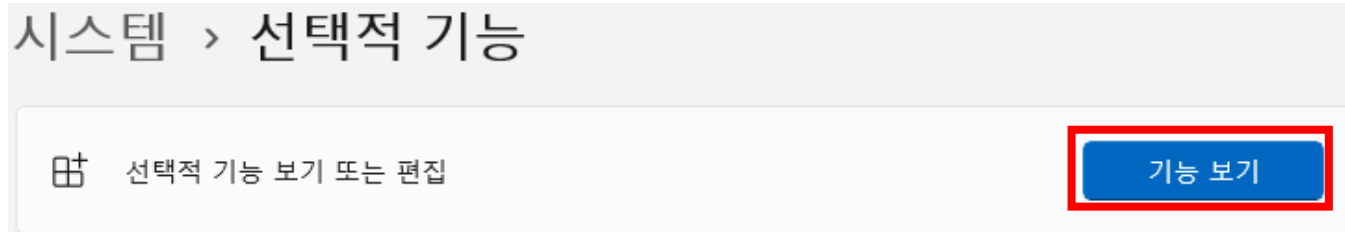


- **Add WMIC (설치되어 있지 않은 경우 만)**
 - FWDN 전 AI-G 보드의 FWDN 포트에 연결된 포트를 알기 위해 WMIC 설치 필요
 - 선택적 기능 메뉴 선택

AI-G 환경 설정

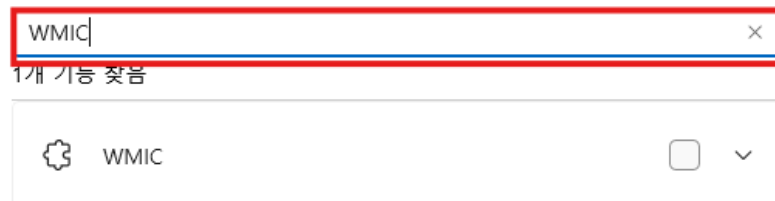
• Add WMIC

- FWDN 전 AI-G 보드의 FWDN 포트에 연결된 포트를 알기 위해 WMIC 설치 필요
 - 기능 보기 선택



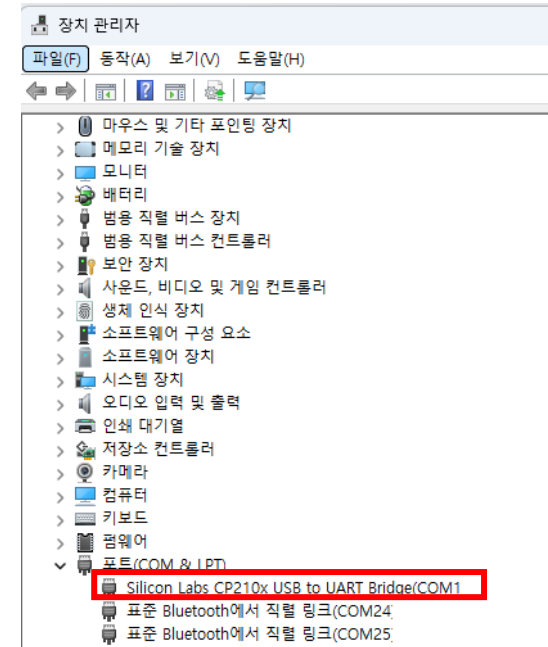
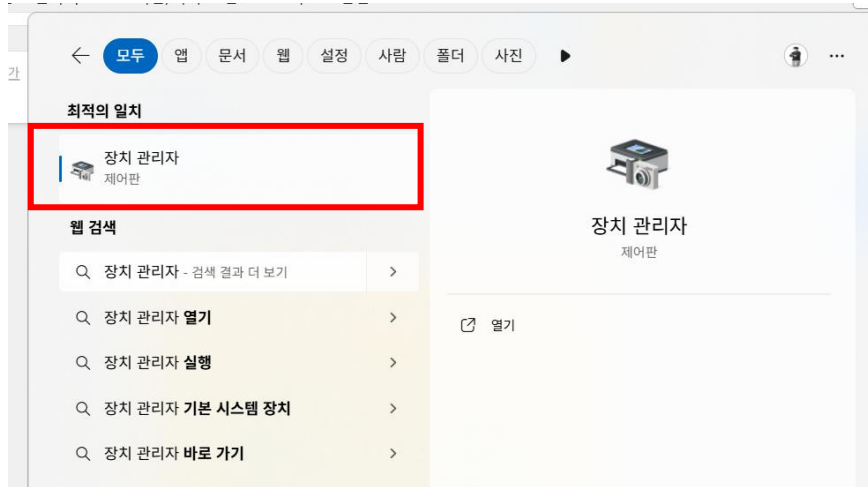
- 검색창에 'WMIC' 입력

선택적 기능 추가



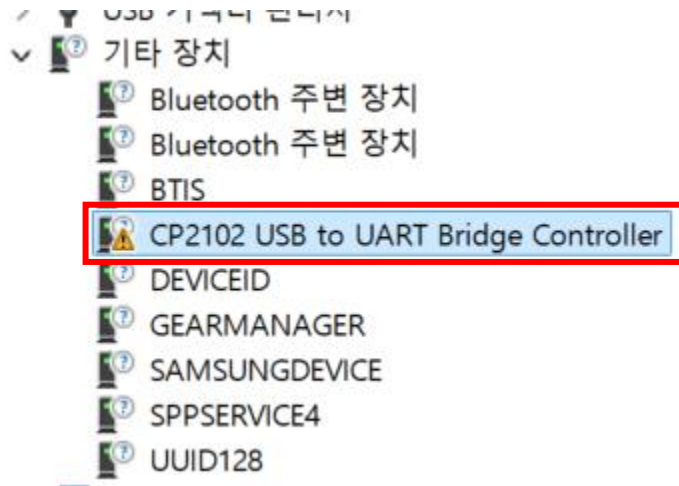
- WMIC를 선택하여 다운로드

AI-G FWDN



- **FWDN (AI-G Yocto Image)**
 - 검색창에 장치 관리자 검색 후 실행
 - 포트 탭에서 연결된 장치 시리얼 번호 (COM xx 확인)

AI-G FWDN

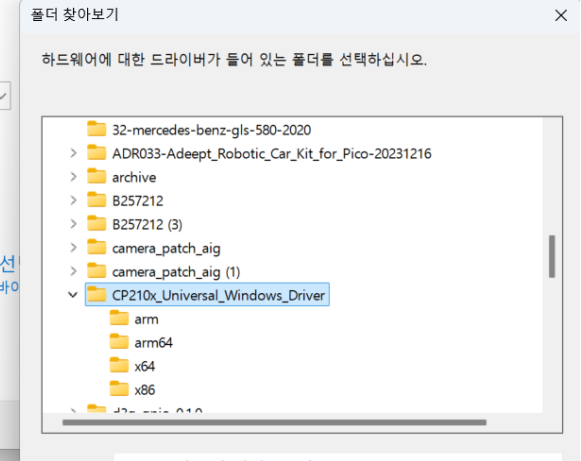


드라이버 찾아보기

검색:

\\OneDrive - Telechips\문서

가능한 드라이버 목록에서 직접 선택하거나, 하위 폴더를 클릭하여 하위 폴더를 선택할 수 있습니다.



• 장치가 보이지 않을 경우

- 기타 장치에서 CP2102 우클릭 -> 드라이버 업데이트 -> 내 컴퓨터에서 드라이버 찾아보기
- 찾아보기 -> 이전에 설치한 CP210x 드라이버 폴더 선택 후 확인 -> 다음
- 장치 인식 확인

AI-G FWDN

```
C:\WINDOWS\system32\cmd.exe
TOPST AI-G FWDN Batch File

Find Port.
Silicon Labs CP210x USB to UART Bridge(COM7)
표준 Bluetooth에서 직렬 링크(COM24)
Bluetooth Device (RFCOMM Protocol TDI)
표준 Bluetooth에서 직렬 링크(COM25)

Input USB Port Number :
```



```
TOPST AI-G FWDN Batch File

Find Port.
Silicon Labs CP210x USB to UART Bridge(COM44)

Input USB Port Number : 44

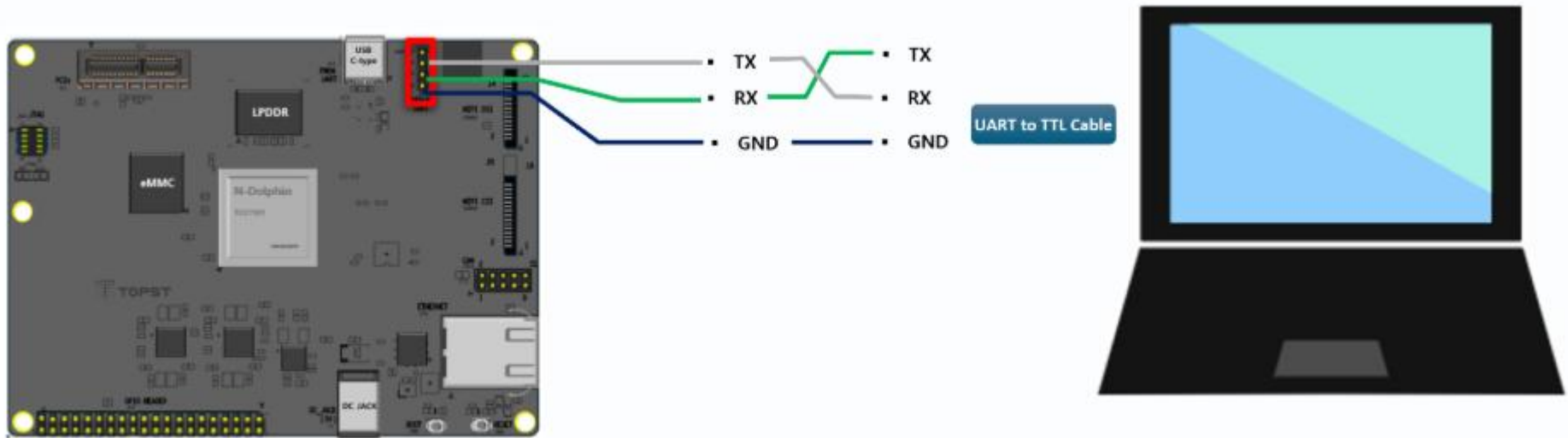
Step 1. Connect between Host PC and TOPST AI-G
[FWDNLogger::PrintCurTime:100] 25/06/12-16:11:36
[main:24] FWDN N-Dolphin v1.0.1 - 2024.9.24 16:18:11
[main:92] Start FWDN
[FWDN_ND::ImageSend:442] Complete to send MCERT image
[FWDN_ND::ImageSend:442] Complete to send HSM image
[FWDN_ND::ImageSend:442] Complete to send FWDN_BL1 image
[FWDN_ND::ImageSend:442] Complete to send FWDN_DRAM_PARAMETER image
[FWDN_ND::ImageSend:442] Complete to send FWDN_BL2 image
[FWDN_ND::ImageSend:442] Complete to send FWDN_BL3 image
[FWDN_ND::InfoDevice:123] ##### Device Info #####
[PrintIP:527] ip : 192.168.0.100
[PrintIP:527] mask : 255.255.255.0
[PrintIP:527] gateway : 0.0.0.0
[FWDN_ND::InfoDevice:132] port : 8080
[FWDN_ND::InfoDevice:135]
mmc0_user: 7818182656 byte block_size: 512 byte
mmc0_boot0: 4194304 byte block_size: 512 byte
mmc0_boot1: 4194304 byte block_size: 512 byte
snor1: 0 byte
snor2: 0 byte
fw_version: tcc750x_v0.0.111
fw_build_id: g90e2b777
fw_build_date: 2024-09-24-17:11:42+0900

----- Firmware Information -----
VERSION : tcc750x_v0.0.111
BUILD ID : g90e2b777
BUILD DATE : 2024-09-24-17:11:42+0900
```

• FWDN 실행

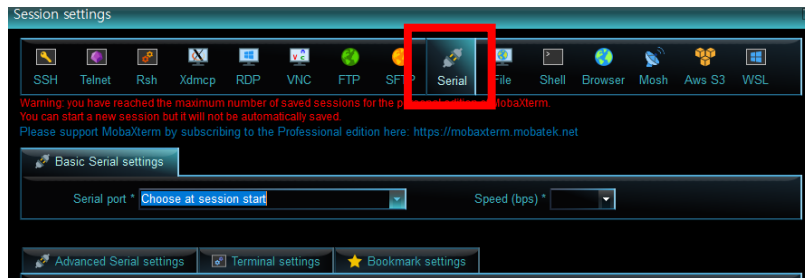
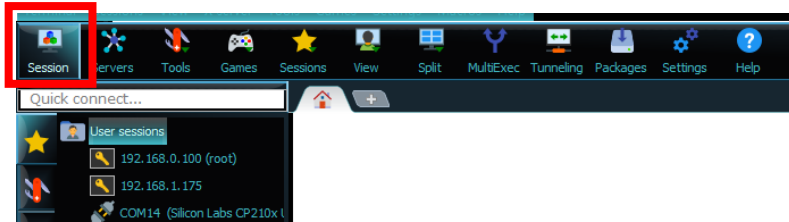
- fwn_aig (windows 배치파일) 실행
- USB 포트 넘버에 확인한 시리얼 장치 번호 입력 - 14

AI-G ↔ PC 통신



- **Connect AI-G Board and Host PC with UART**
 - AI-G 보드와 PC 간 USB-to-TTL cable을 이용하여 연결
 - 검정 케이블은 UART 핀의 GND에 연결
 - 흰 케이블은 UART 핀의 TX 핀에 연결 & 녹색 케이블은 UART 핀의 RX 핀에 연결

AI-G ↔ PC 통신



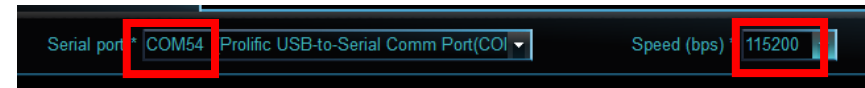
• Connect AI-G Board and Host PC with UAR

- 보드 접속
 - MobaXterm 실행
 - 장치 관리자를 통해 port 확인
 - ✓ 포트(COM & LPT)
 - Prolific USB-to-Serial Comm Port(COM36)
 - Silicon Labs CP210x USB to UART Bridge(COM4)
 - Session -> Serial 탭 클릭
 - Serial port 탭에서 장치관리자를 통해 확인한 포트 선택
 - Speed '115200'으로 설정

AI-G ↔ PC 통신



- Serial port에서 연결된 포트 설정
- Speed 115200 설정



- Advanced Serial settings 탭 선택
- Flow contro에서 None 선택
- OK 눌러 보드 접속

AI-G ↔ PC 통신

```
eth0: gmac@13000000
blkread: mmc 0 is current device
Hit any key to stop autoboot: 0
[tcc_fb_dm]: version : 1.4.0 20230824
Video      = tcc_fb_dm0 active
Non-secure boot (secure boot flag is clear)
## Booting Android Image at 0x24000000 ...
Kernel load addr 0x20000000 size 16517 KiB
Kernel command line: quiet fsck.repair=yes console=ttyAMA2,115200n8
## Flattened Device Tree blob at 28000000
   Booting using the fdt blob at 0x28000000
   Loading Kernel Image
ERROR: reserving fdt memory region failed (addr=20000000 size=1000000 flags=4)
   Using Device Tree in place at 0000000028000000, end 0000000028010fff
   DTB 05 may not be stored in storage. res: 0xffffffffffffffff

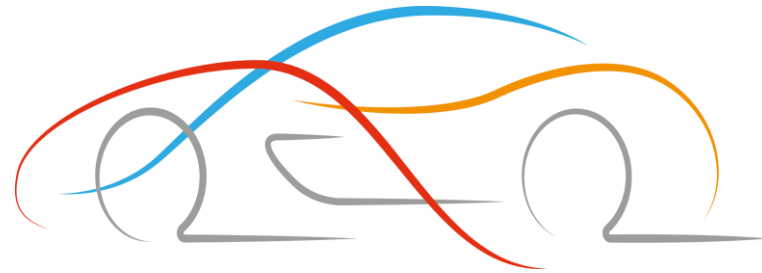
Starting kernel ...

[ 1.207999] telechips-pcie 12000000.pcie: invalid resource
[ 1.334582] telechips-pcie 12000000.pcie: timeout waiting for link up
[ 1.592471] debugfs: Directory '18300000.dma' with parent 'dmaengine' already present!
[ 1.601609] debugfs: Directory '18320000.dma' with parent 'dmaengine' already present!
[ 1.610745] debugfs: Directory '18330000.dma' with parent 'dmaengine' already present!
[ 1.619885] debugfs: Directory '18340000.dma' with parent 'dmaengine' already present!
[ 1.629051] debugfs: Directory '18340000.dma' with parent 'dmaengine' already present!
[ 4.990360] tcc-dewarp videoinput0: [ERROR][DEWARP] tcc_dewarp_ioctl_enum_framesizes - index(1) is not 0
[ 4.999920] tcc-dewarp videoinput0: [ERROR][DEWARP] tcc_dewarp_ioctl_enum_framesizes - index(1) is not 0
[ 5.000469] tcc-dewarp videoinput0: [ERROR][DEWARP] tcc_dewarp_ioctl_enum_framesizes - index(1) is not 0
[ 5.019084] tcc-dewarp videoinput0: [ERROR][DEWARP] tcc_dewarp_ioctl_enum_framesizes - index(1) is not 0
[ 5.028523] tcc-dewarp videoinput0: [ERROR][DEWARP] tcc_dewarp_ioctl_enum_framesizes - index(1) is not 0
[ 5.038059] tcc-dewarp videoinput0: [ERROR][DEWARP] tcc_dewarp_ioctl_enum_framesizes - index(1) is not 0
[ 5.047577] tcc-dewarp videoinput0: [ERROR][DEWARP] tcc_dewarp_ioctl_enum_framesizes - index(1) is not 0
[ 5.057141] tcc-dewarp videoinput0: [ERROR][DEWARP] tcc_dewarp_ioctl_enum_framesizes - index(1) is not 0
[ 5.057155] tcc-dewarp videoinput0: [ERROR][DEWARP] tcc_dewarp_ioctl_enum_framesizes - index(1) is not 0
[ 5.076165] tcc-dewarp videoinput0: [ERROR][DEWARP] tcc_dewarp_ioctl_enum_framesizes - index(1) is not 0

Telechips Baseline (Poky/meta-telechips/meta-core) 4.0.17 ai-g-topst ttyAMA2
ai-g-topst login: |
```

```
root@ai-g-topst:~# _
```

- 보드 실행
 - Reset 버튼을 누르거나 전원 재인가
 - 디버깅 창 확인
 - ai-g-topst login : root / password : root 입력



Thank You